

CHAPTER - 1 : INTRODUCTION TO DATA SCIENCE

* 1. What is Data ?

Data is a collection of **raw facts** and figures that have no meaning until processed.

Example : • 85, 90, 78

- Temperature = 35°C
- Name = Rahul

* 2. What is Information ?

Information is **processed data** that has **meaning** and helps in decision-making.

Example :

Data → 85, 90, 78

Information → Average Marks = 84.3

3. What is Data Science ?

Data Science is the process of **collecting**, **cleaning**, **analyzing** and **visualizing** data to find useful information and make better decisions.

Formula :

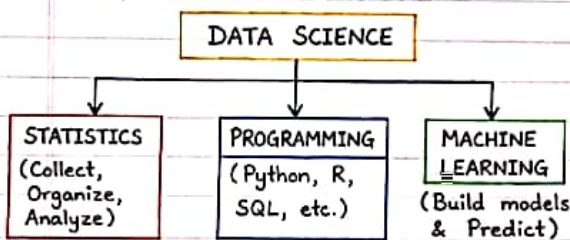
Data Science = Data + Statistics
+ Programming + Machine Learning

4. Why Data Science ?

Data Science helps to :

- Predict future trends
- Make better business decisions
- Find hidden patterns
- Automate tasks
- Solve real-world problems

5. Components of Data Science



7. Types of Data

Structured	Semi-Structured	Unstructured
• Stored in rows and columns. • Easy to search and analyze. Example : Excel, MySQL, Student DB	• Does not follow fixed table. • Uses tags or markers. Example : XML, JSON	• No predefined format. • Difficult to analyze. Example : Images, Videos, Audio, Text

8. Applications of Data Science

- Healthcare
- Banking & Finance
- Agriculture
- Education
- E-commerce
- Weather Forecasting
- Social Media
- Cyber Security
- Sports Analytics
- Transportation

6. Data Science Life Cycle

- 1 Problem Definition → Understand the problem.
- 2 Data Collection → Gather data from different sources.
- 3 Data Cleaning → Remove errors, missing values and noise.
- 4 Data Analysis → Analyze data to find useful insights.
- 5 Model Building → Train machine learning model.
- 6 Evaluation → Check model accuracy and performance.
- 7 Deployment → Use the model in real world applications.

9. Advantages

- Better Decision Making
- Accurate Prediction
- Saves Time
- Business Growth
- Automation
- Improve Customer Experience

10. Disadvantages

- High Cost
- Requires Skilled People
- Privacy Issues
- Large Storage Required
- Data Security Risks

11. Important Python Libraries in Data Science

Library	Use	Library	Use
1. NumPy	Numerical calculations	4. Seaborn	Advanced statistical visualization
2. Pandas	Data manipulation & analysis	5. Scikit-learn	Machine Learning algorithms
3. Matplotlib	Data visualization (plots, graphs)		

Exam Tip

Learn diagrams, definitions and examples well. They are commonly asked in MSBTE exams.



12. Frequently Asked Questions (Viva)

Q1. What is Data Science ?

Ans: Data Science is the process of extracting useful information from data using statistics, programming and machine learning.

Q2. Difference between Data and Information ?

Ans: Data → Raw facts
Information → Processed data with meaning

Q3. Name any four Python libraries used in Data Science.

Ans: NumPy, Pandas, Matplotlib, Scikit-learn.

Q4. What is Machine Learning ?

Ans: Machine Learning is a branch of AI that enables computers to learn from data without being explicitly programmed.

1. Data Collection

Data collection is the process of **gathering** data from **different sources**.

Methods :

- ① Surveys & Questionnaires
- ② Interviews
- ③ Observation
- ④ Experiments
- ⑤ Web Scraping
- ⑥ Sensors & IoT Devices
- ⑦ Existing Datasets (Kaggle, UCI, etc.)

Good Quality
Data
leads to
Good Results

2. Data Preprocessing

Raw data is not clean. We need to convert it into **clean** and **useful** data.

Steps :

- ① Data Cleaning
- ② Handling Missing Values
- ③ Removing Duplicates
- ④ Data Transformation
- ⑤ Data Normalization
- ⑥ Data Reduction

Why Preprocess?

- Improves accuracy
- Removes noise
- Better performance
- Easy to analyze

3. Handling Missing Values

Method	Description
Drop Missing	Remove rows/columns with missing values.
Mean/Median	Replace missing values with mean or median.
Mode	Replace with most frequent value.
Forward Fill	Use previous value to fill.
Backward Fill	Use next value to fill.

Missing
Values

5. Exploratory Data Analysis (EDA)

EDA is used to analyze and summarize the main characteristics of data using **visual** and **statistical** methods.

Goals :

- Understand the data
- Detect patterns
- Find relationships
- Detect outliers
- Test hypothesis

EDA helps
in better
decision
making.



4. Data Cleaning

- Remove duplicate data
- Fix incorrect data
- Handle outliers
- Standardize formats
- Check data consistency

Example :

Before Cleaning

85, 90, 90, 120, -10, NA



After Cleaning

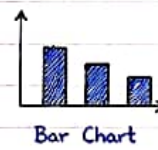
85, 90, 90, 95



6. Data Visualization

Data visualization represents data in **graphical** format to understand it **easily**.

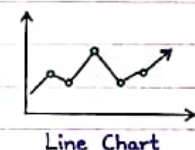
Common Plots :



Bar Chart



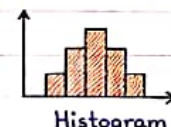
Pie Chart



Line Chart



Scatter Plot



Histogram

7. Types of Machine Learning

Supervised Learning	Unsupervised Learning	Reinforcement Learning
• Uses labeled data • Has input & output • Model learns mapping • Examples : Regression, Classification	• Uses unlabeled data • Finds hidden patterns • No output provided • Examples : Clustering, Association	• Learns by trial & error • Uses rewards & penalties • Examples : Game Playing, Robotics

Difference ☆

Type	Data	Goal
Supervised	Labeled	Predict Output
Unsupervised	Unlabeled	Find Patterns
Reinforcement	Feedback	Maximize Reward

8. Popular Algorithms

Algorithm	Use
Linear Regression	Predict continuous values
Logistic Regression	Classification (Yes/No)
Decision Tree	Classification & Regression
K-Nearest Neighbors (KNN)	Classification
K-Means Clustering	Clustering / Grouping
Naive Bayes	Text Classification
Support Vector Machine (SVM)	Classification

Algorithm
selection
depends on
problem type
and data.

9. Evaluation Metrics

Metric	Use
Accuracy	$(TP + TN) / \text{Total}$
Precision	$TP / (TP + FP)$
Recall (Sensitivity)	$TP / (TP + FN)$
F1-Score	$2 \times \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right)$
Confusion Matrix	Shows TP, TN, FP, FN

Important Points to Remember ☆

- * Data is the new oil.
- * Quality data gives quality insights.
- * Always clean your data before analysis.

- * Visualization makes data easy to understand.
- * Right algorithm + Right data = Right results.

Practice
Makes
Perfect

1. TensorFlow - Introduction

TensorFlow is an open source machine learning library developed by Google.

It is used to build and train machine learning and deep learning models.

Key Features :

- Easy to use
- Supports deep learning
- Runs on CPU, GPU, TPU
- Scalable and flexible
- Large community support



2. Uses of TensorFlow

- Image recognition
- Natural Language Processing (NLP)
- Speech recognition
- Time series forecasting
- Recommendation systems
- Computer vision applications

5. Descriptive Statistics

Descriptive statistics describes and summarizes the main features of data.

Measure	Formula	Use
Mean (Average)	$\bar{x} = \frac{\sum x}{n}$	Central value of data
Median	Middle value	When data is in order
Mode	Most frequent value	For categorical data
Range	Max - Min	Spread of data
Variance (σ^2)	$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$	Measures dispersion
Standard Deviation (σ)	$\sigma = \sqrt{\sigma^2}$	Square root of variance

Note :
Mean is affected by extreme values, Median is not.

7. Measures of Central Tendency

Mean	Median	Mode
• Uses all values • Best for normal distribution • Affected by outliers	• Middle value • Best for skewed data • Not affected by outliers	• Most occurring value • Used for categorical data

For Normal Distribution
Mean = Median = Mode

9. Probability Basics

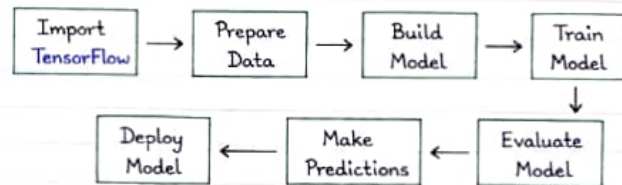
- Probability is the chance of an event to occur.
- Range : $0 \leq P(E) \leq 1$
- Formula : $P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$

11. Correlation and Covariance

- **Correlation (r)** : Measures strength and direction of relationship between two variables.
 $-1 \leq r \leq 1$
- **Covariance (Cov)** : Measures how two variables change together.

Value of r	Interpretation
$r = +1$	Perfect positive correlation
$0 < r < +1$	Positive correlation
$r = 0$	No correlation
$-1 < r < 0$	Negative correlation
$r = -1$	Perfect negative correlation

3. Basic TensorFlow Workflow



4. Simple Example - Using TensorFlow

```

import tensorflow as tf
import numpy as np

x = np.array([1,2,3,4,5,6], dtype=float)
y = np.array([2,4,6,8,10,12], dtype=float)

model = tf.keras.Sequential([
    tf.keras.layers.Dense(1, input_shape=(1,))
])

model.compile(optimizer='sgd', loss='mse')
model.fit(x, y, epochs=1000, verbose=0)
print("Prediction for 7 =", model.predict([7]))
    
```

This program trains a model to learn relation between x and y.

6. Frequency Distribution Table (Example)

Marks of 30 students out of 100

Class Interval	Frequency (f)	Cumulative Freq. (cf)
0 - 20	2	2
20 - 40	5	7
40 - 60	8	15
60 - 80	10	25
80 - 100	5	30

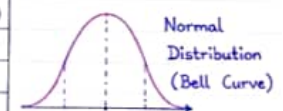
cf = Sum of all previous frequencies + current frequency.

8. Measures of Dispersion

Measure	Formula	Use
Range	Max - Min	Simple measure of spread
Variance (σ^2)	$\frac{\sum (x - \bar{x})^2}{n}$	Average squared deviation
Standard Deviation (σ)	$\sqrt{\frac{\sum (x - \bar{x})^2}{n}}$	Most widely used measure
Coefficient of Variation (CV)	$\left(\frac{\sigma}{\bar{x}}\right) \times 100\%$	Compare variation between datasets

10. Common Probability Distributions

Distribution	Use
Binomial Distribution	Two possible outcomes (Success/Failure)
Normal Distribution	Continuous data, bell shaped curve
Poisson Distribution	Count of events in fixed interval
Uniform Distribution	All outcomes are equally likely



12. Important Statistical Concepts

Population : Entire set of elements.
Sample : Subset of population.
Parameter : Measure about population (μ, σ)
Statistic : Measure about sample ($\bar{x}, s$)
Hypothesis : Assumption about population.
Null Hypothesis (H_0) : No effect or no difference.
Alternative Hypothesis (H_1) : There is effect or difference.

Statistics helps us to make decisions based on data.

Important Formulas (Quick Revision)

- Mean ($\bar{x}$) = $\frac{\sum x}{n}$
- Variance (σ^2) = $\frac{\sum (x - \bar{x})^2}{n}$
- Standard Deviation (σ) = $\sqrt{\frac{\sum (x - \bar{x})^2}{n}}$

- Probability
- Covariance
- Correlation

$$P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$$

$$\text{Cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n}$$

$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

Remember

Collect → Clean → Analyze
→ Model → Evaluate
→ Deploy

1. Hypothesis Testing

Hypothesis testing is a statistical method used to make a decision about a population parameter based on sample data.

Steps :

- State the null hypothesis (H_0)
- State the alternative hypothesis (H_1)
- Choose level of significance (α)
- Calculate test statistic
- Take decision (Reject or Accept H_0)

If p-value
< α
Reject H_0
else
Accept H_0
😊

H_0 (Null Hypothesis) : There is no effect or no difference.

H_1 (Alternative Hypothesis) : There is an effect or difference.

3. ANOVA (Analysis of Variance)

ANOVA is used to find whether there is a significant difference between the means of three or more groups.

ANOVA Table (Example)

Source	SS	df	MS	F
Between Groups	SS _B	k-1	MS _B	$\frac{MS_B}{MS_W}$
Within Groups	SS _W	N-k	MS _W	-
Total	SST	N-1	-	-

k = number
of groups
N = total
observations
😊

If $F_{cal} > F_{critical} \rightarrow$ Reject H_0

5. Classification and Clustering

Classification (Supervised)	Clustering (Unsupervised)
- Used when output is categorical. - Example Algorithms : - Logistic Regression - Decision Tree - Support Vector Machine - Naive Bayes - Example : Spam Detection, Disease Prediction	- Used when output is not known. - Example Algorithms : - K-Means Clustering - Hierarchical Clustering - DBSCAN - Example : Customer Segmentation, Market Basket Analysis

7. Big Data - 5 Vs

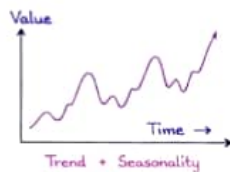


9. Time Series Analysis

Time series data is collected at different points in time.

Components :

- Trend - Long term movement
- Seasonality - Repeating pattern
- Cyclic - Irregular cycles
- Irregular / Noise - Random variation



11. Sampling Techniques

Type	Description
Simple Random Sampling	Every member has equal chance
Systematic Sampling	Select every k-th item
Stratified Sampling	Divide into strata and sample
Cluster Sampling	Divide into clusters and sample
Convenience Sampling	Select data that is easy to collect
Purposive Sampling	Select based on judgement

Why Sampling ?

- Saves Time
- Reduces Cost
- Easy to Manage
- Gives Faster Results

2. Types of Hypothesis Testing

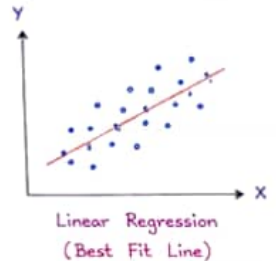
Test	Used For	Example
Z-Test	Large sample ($n \geq 30$) Population σ known	Testing mean height
t-Test	Small sample ($n < 30$) Population σ unknown	Testing average marks
Chi-Square Test	Categorical data	Goodness of fit test Independence test
F-Test (ANOVA)	Compare more than two groups	Compare average sales of 3 different regions

4. Regression Analysis

Regression is used to find the relationship between dependent and independent variables.

Types :

- Linear Regression $\rightarrow Y = a + bX$
- Multiple Regression $\rightarrow Y = a + b_1X_1 + b_2X_2 + \dots + b_nX_n$



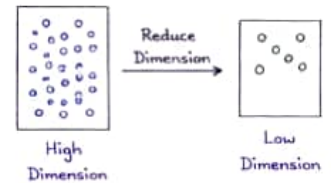
Y = Dependent Variable
X = Independent Variable
a = Intercept
b = Slope (Regression Coefficient)

6. Dimensionality Reduction

It reduces the number of input variables while retaining important information.

Methods :

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- t-SNE
- Autoencoders



8. Data Mining Tasks

- Association Rule Mining - Find relationships (Market Basket Analysis)
- Sequence Mining - Find ordered patterns
- Classification - Predict category
- Clustering - Group similar data
- Regression - Predict continuous values
- Outlier Detection - Find unusual data

Example - Association Rule

If a customer buys {Bread} then he is likely to buy {Butter}

Support = 30%

Confidence = 70%

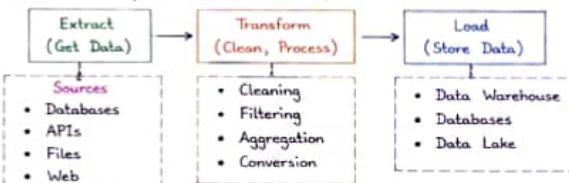
Lift = 1.4

10. Data Sources

Primary Data	Secondary Data
- Surveys - Questionnaires - Interviews - Experiments - Observations	- Books - Research Papers - Websites - Government Reports - Databases

Good data
is the
foundation
of good
analysis.
😊

12. Data Pipeline (ETL Process)



13. Advanced Analytics

- Predictive Analytics - Predict future outcomes
- Prescriptive Analytics - Suggest best actions
- Diagnostic Analytics - Find cause of events
- Descriptive Analytics - Summarize past data

Quick Formulae ☆

$$\text{Mean } (\bar{x}) = \frac{\sum x}{n}$$

$$\text{Variance } (\sigma^2) = \frac{\sum (x - \bar{x})^2}{n}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

14. Important Libraries (Python)

Library	Use
NumPy	Numerical operations
Pandas	Data manipulation & analysis
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Machine Learning
Statsmodels	Statistical modeling
Plotly	Interactive plots
TensorFlow / Keras	Deep Learning

15. Real World Applications

- Healthcare - Disease prediction
- Finance - Fraud detection
- Retail - Customer segmentation
- Manufacturing - Quality control
- Education - Student performance
- Agriculture - Crop prediction
- Transportation - Demand forecasting

Remember

Collect $\rightarrow$ Clean $\rightarrow$
Analyze $\rightarrow$ Interpret $\rightarrow$
Decide $\rightarrow$ Act
😊

Exam Tip $\rightarrow$ Draw diagrams, tables and write short